Seasonal Dynamics and Thermal Vulnerability of Marine Microbial Communities in a Coastal South Atlantic Warming Hotspot

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Introduction

- In an era of accelerating ocean warming, understanding microbial community dynamics is more urgent than ever.
- Long-term time-series studies are particularly powerful for this purpose. However, most previous works remain confined to the Northern Hemisphere and focus solely on taxonomy, leaving the Southern Hemisphere underrepresented and microorganisms' functional dimensions underexplored.
- In addition, current-driven shifts and microbial vulnerability to warming are also rarely examined. Expanding this knowledge is vital for modeling and mitigating impacts on ecosystems and biogeochemical cycles.
- Here, we analysed two years of microbial communities at the South Atlantic Microbial Observatory (SAMO) using amplicon and metagenomic sequencing. SAMO, off Uruguay within a Marine Protected Area and a warming hotspot, offers an ideal setting.

Results

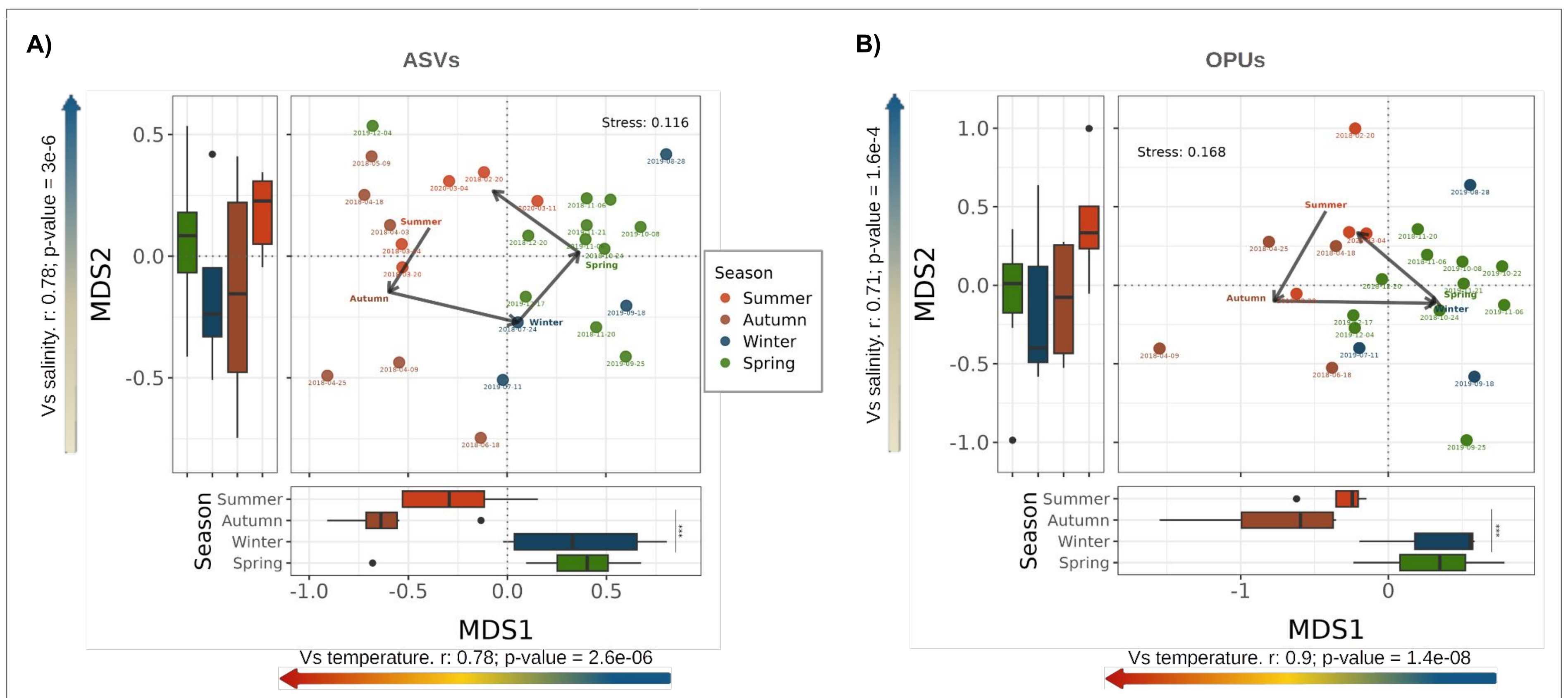

Figure 1. Community differentiation linked to seasonal and ocean current dynamics. A and B) Non-Metric Multidimensional Scaling (NMDS) of Bray-Curtis dissimilarity matrices from Amplicon Sequence Variants (ASVs) and Operational Protein Units (OPUs). Samples are colour coded by season. Arrows connect seasonal centroids, showing transitions over time. Boxplots display MDS1 and MDS2 distributions across seasons. In both analyses, MDS1 separated summer/autumn from winter/spring (Welch ANOVA, p < 0.001). Coloured arrows represent temperature and salinity, strongly correlated with MDS1 and MDS2, respectively.

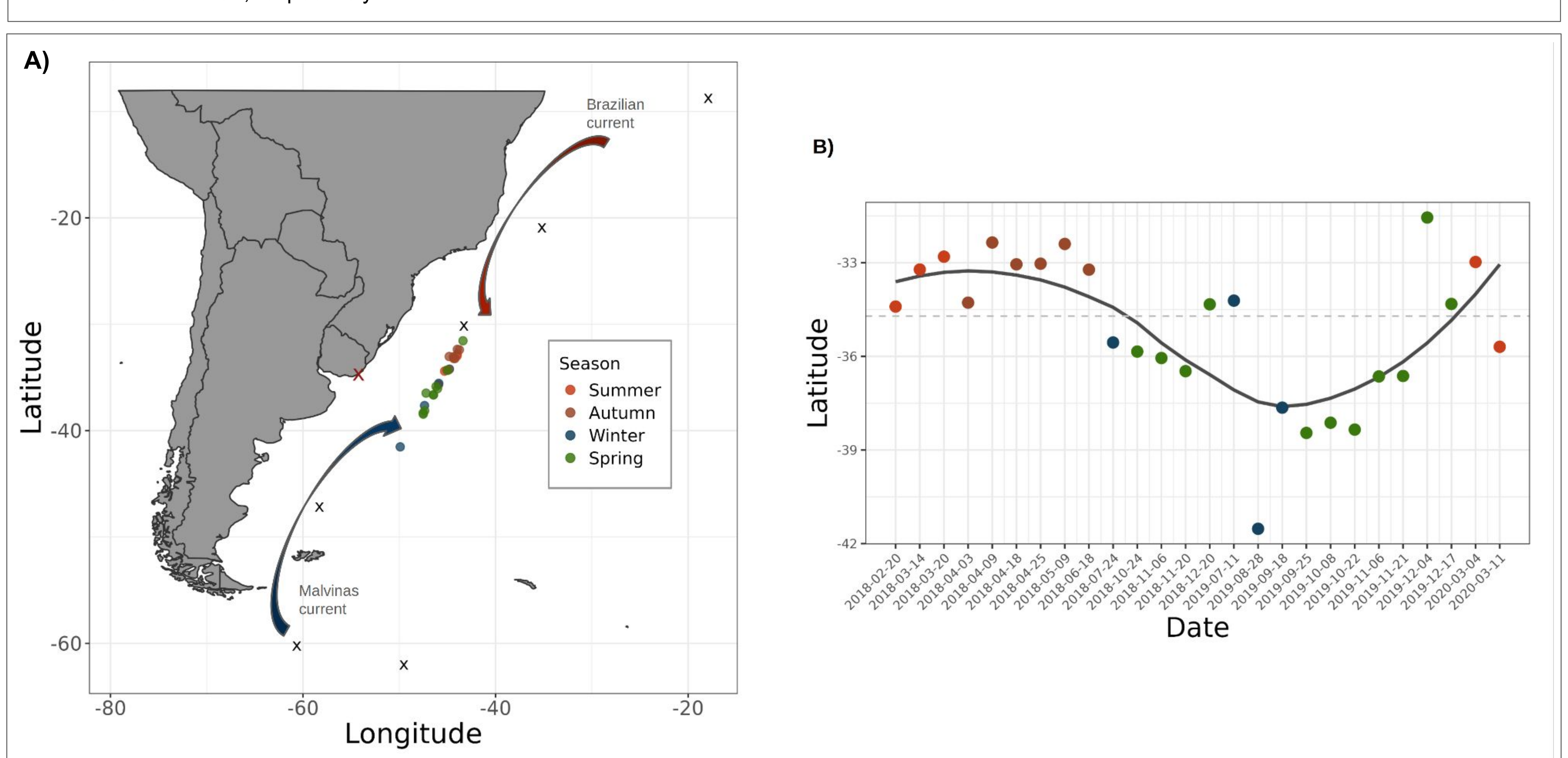

Figure 2. Community differentiation linked to ocean current dynamics. A) Space-time analysis of genus-level composition comparing SAMO with TARA Oceans (1). SAMO samples were positioned relative to six South Atlantic TARA sites using Bray-Curtis similarity of genus profiles. SAMO coordinates were calculated as weighted averages of TARA locations, weighted by Bray-Curtis similarity. Black crosses mark TARA sites, the red cross marks SAMO, colours indicate seasons. Brazil and Malvinas currents are shown schematically. **B)** Scatterplot showing variability of the imputed latitude coordinate over time, with a LOESS regression line.

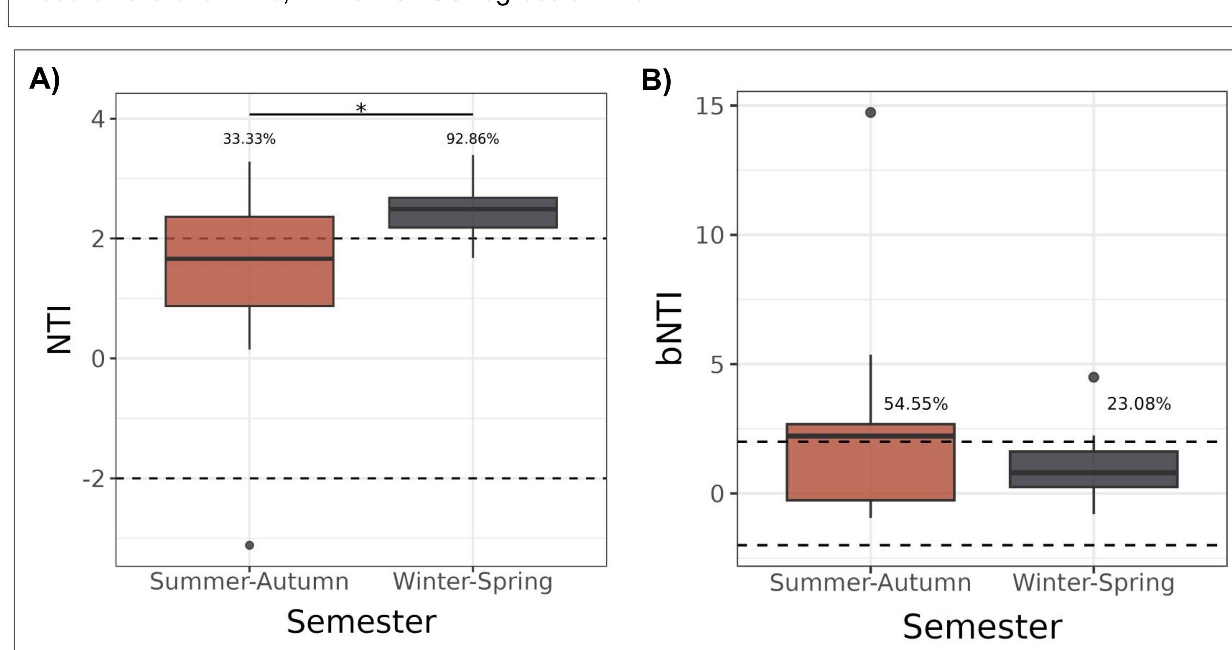

Figure 3. Ecological processes shaping community composition. A and B) Boxplots of Nearest Taxon Index (NTI) and beta NTI (bNTI) (2) for Summer/Autumn and Winter/Spring samples. NTI and bNTI medians were significantly > 0 (Wilcoxon Signed Rank Sum; p < 0.05). Winter/Spring NTI values were higher than Summer/Autumn (Welch's ANOVA, p = 0.035), while bNTI showed no difference (p = 0.28). Percentages above boxes indicate samples with NTI > 2.

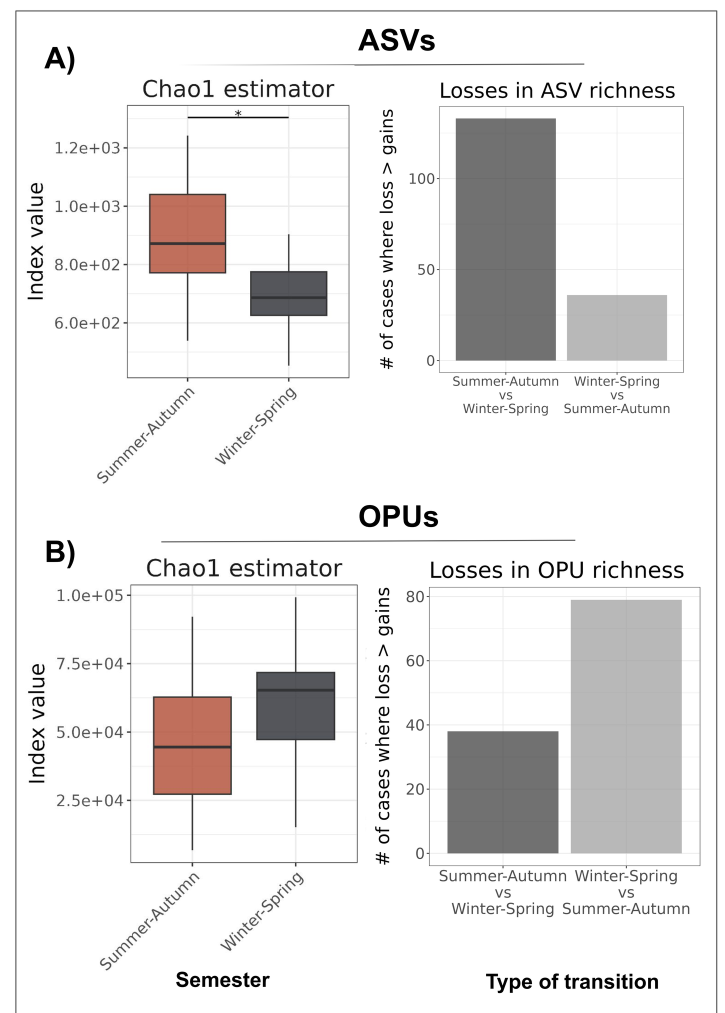

Figure 4. Contrasting diversity patterns between semesters. A and B) Boxplots comparing Chao1 estimators of microbial communities from Summer-Autumn and Winter-Spring at ASV and OPU levels, respectively. Losses in richness refer to decreases in ASVs or OPUs when transitioning between semesters (3).

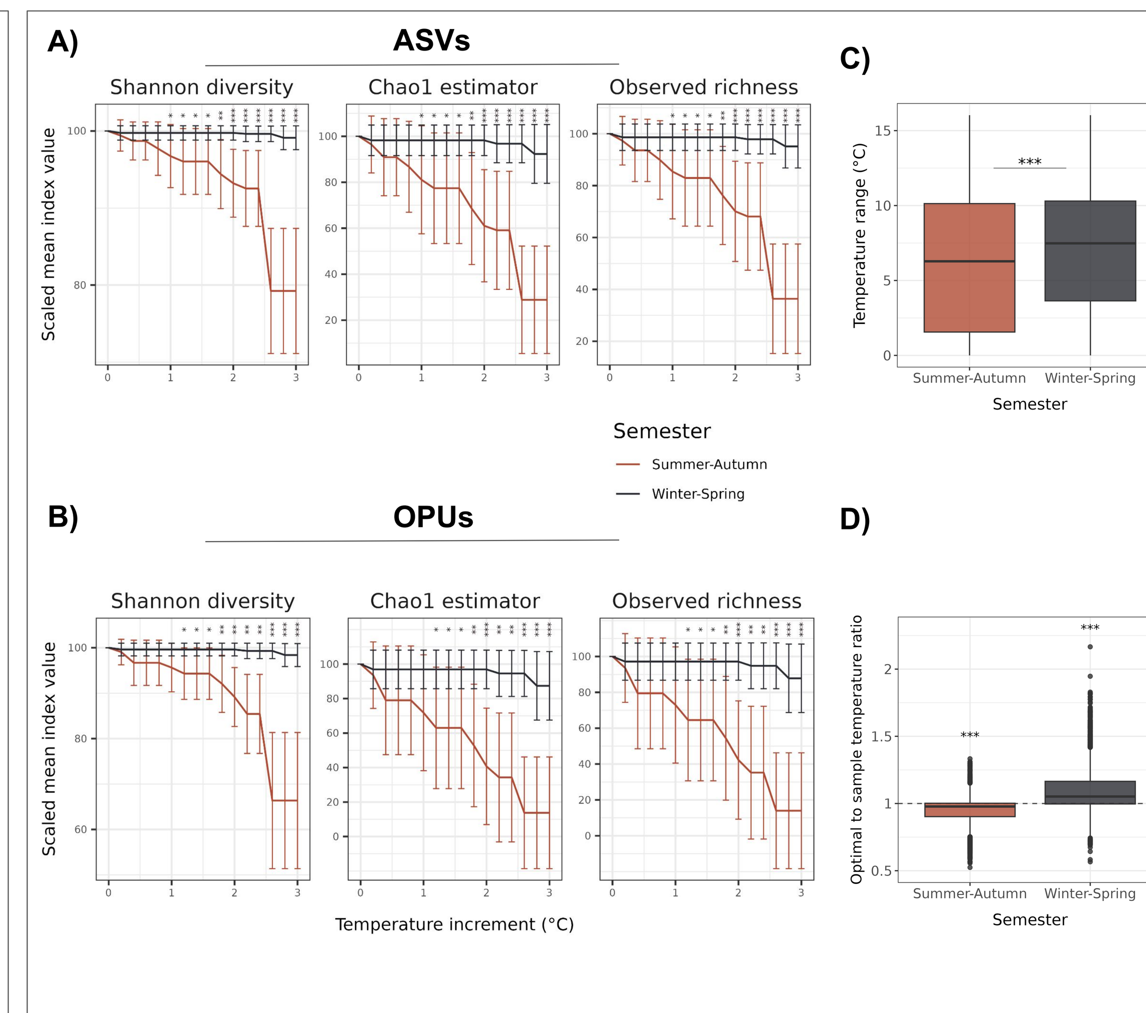

Figure 5. Simulation of increased water temperature scenarios. A and B) Simulated temperature increases (0–3 °C) on ASV and OPU community compositions evaluated through the diversity indices (Fig. 7). Semester differences were tested with Welch's ANOVA. **C)** Boxplots show broader ASV temperature ranges in Winter/Spring than Summer/Autumn (Welch's ANOVA, p < 2.2e-16). **D)** Boxplots of the ratio between ASV optimal temperature (abundance-weighted mean) and in situ temperature. Ratios fall below 1 in Summer/Autumn and above 1 in Winter/Spring (Wilcoxon, p < 0.001), as marked by asterisks.

Take-home messages

1. **Marine microbial communities off Uruguay's coast form two recurring seasonal assemblages (Summer/Autumn and Winter/Spring), strongly shaped by deterministic processes.**
2. **Their dynamics track shifts in temperature, salinity, and the Brazil/Malvinas currents, with communities adapted to oligotrophic conditions in Summer–Autumn and nutrient-rich (copiotrophic) conditions in Winter–Spring.**
3. **Communities from different semesters show contrasting diversity patterns, implying losses of functional or taxonomic diversity when transitioning from one semester to another.**
4. **The predicted tropicalization of Uruguay's coast may boost taxonomic richness but erode functional repertoires and heighten vulnerability to warming, potentially undermining biogeochemical functioning. This likely reflects tropical communities with narrower thermal windows enduring temperatures above their optimum.**

Methods

1) Samples: 26 amplicon and 22 metagenomes; 2) ASVs: constructed with DADA2 (4); 3) OPUs: clustered with Mg-Clust (70% identity); 4) Normalization: matrices rarefied to minimum depth; 5) Assembly processes: NTI & bNTI (Stegen 2012); 6) Turnover: gains/losses via Bray–Curtis decomposition (Legendre 2019); 7) Thermal stress test: simulated warming scenarios (Fig. 7).

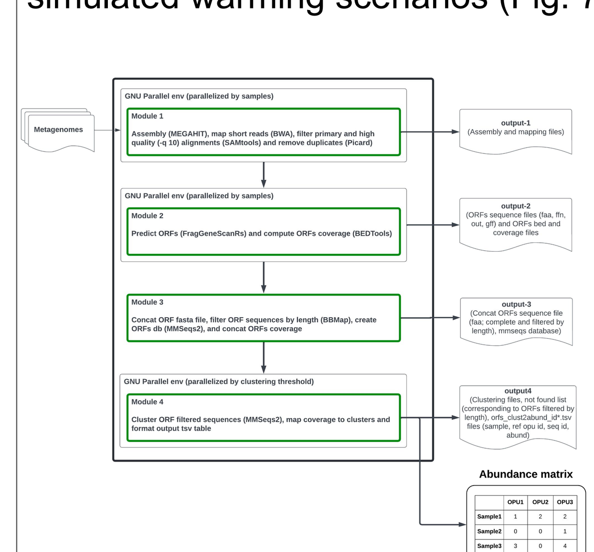

Figure 6. Graphical representation of the Mg-Clust pipeline. The four modules that comprise the pipeline are represented in different boxes, indicating the tasks performed, the tools, and the output files.

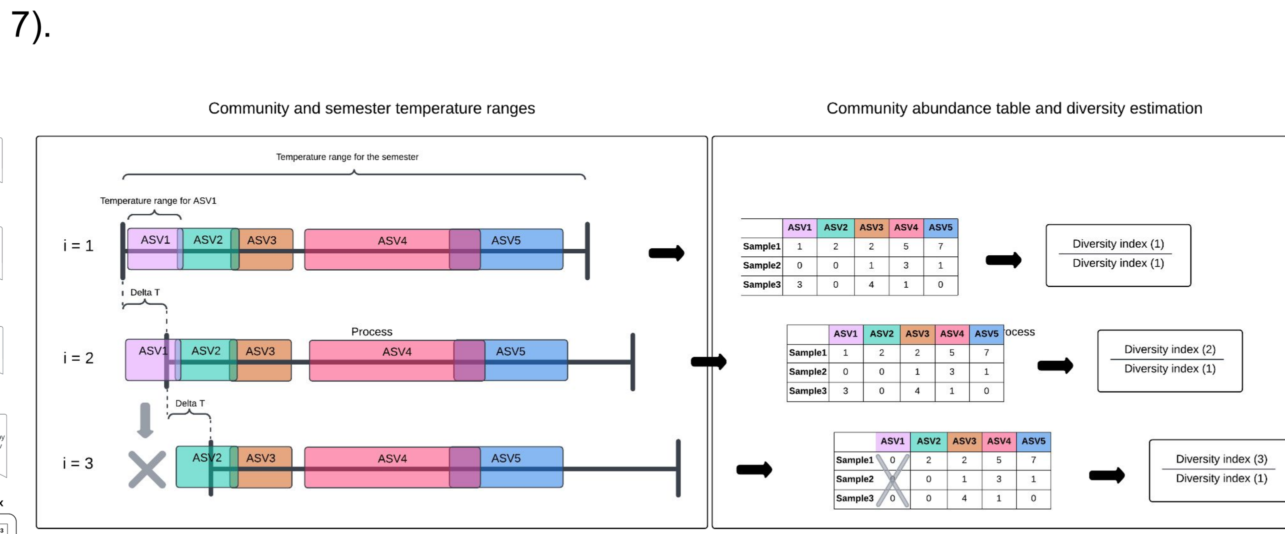

Figure 7. Method for simulating warming scenarios. Three steps with five ASVs are shown. The left panel shows ASV tolerance ranges (boxes) and semester ranges (whiskers). Each iteration shifts the semester range upward by 0.2 °C to 3 °C. ASVs with maxima below the new minimum are removed (abundance = 0). The right panel shows abundance matrices and diversity indices (Shannon, richness, Chao1), scaled to the original. In this example, diversity holds through step 2 but drops in step 3 as ASV1 is excluded.

Bibliography

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